ECN 593 · APPLIED PROJECT SHORT COURSE · LECTURE 4

Two Foils

SUTVA estimators and exposure-mapped inference

ECN 593 short course notes (study draft, not a frozen specification)
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Abstract

The project kernel contains a peer term γ by construction, and the parameter box excludes γ = 0. Individual contrasts of the form $Y_i(1) - Y_i(0)$ are then not well-defined objects: unit i's outcome depends on the entire assignment vector. The well-posed quantities are policy contrasts — θ_1 (random seed versus control), θ_2 (hub seed versus control, the headline), and $\tau = \theta_2 - \theta_1$ (the targeting premium). This lecture develops two diagnostic estimators that a referee will ask for. Neither consumes the not-ruled-out-yet (NROY) set. Foil 1 is a causal forest / Bayesian causal forest fit on a single world c_0; it is a leading method for heterogeneous individual effects under the stable unit treatment value assumption (SUTVA), and it is not an interference estimator. Foil 2, proposed in a draft amendment, is an Aronow–Samii exposure-mapped Hájek estimator that can express a spillover contrast, but only on the randomized arm π_1. That π_2 is deterministic given the graph is a result, not a defect to hide: no design-based estimator reaches the headline policy. History-matching waves have not been run. No numerical comparison is reported here.

1. When $Y_i(1) - Y_i(0)$ is not a defined contrast

The Neyman–Rubin model writes a causal effect as a comparison of potential outcomes on the same unit. In the binary-treatment, no-interference case one writes $Y_i(1)$ and $Y_i(0)$ for the outcomes that unit i would exhibit under treatment and control, and the individual effect is their difference. Holland (1986) named the obstacle to observing both coordinates on the same unit the *fundamental problem of causal inference*. That problem is about missing data. Interference is a different problem. It is about the *index set* of the potential-outcome function.

Cox (1958) required that treatments applied to one unit not affect the outcomes of other units. Rubin (1980, 1986) packaged that requirement, together with the rule that each treatment have a single well-defined version, as the stable unit treatment value assumption (SUTVA). Under SUTVA the map from assignments to outcomes factors: $Y_i(\mathbf{D}) = Y_i(D_i)$. Drop SUTVA and the potential outcome of i is a function of the full vector $\mathbf{D}$ in $\{0,1\}^N$. There are then 2^N potential outcomes per unit, not two. The symbol $Y_i(1) - Y_i(0)$ does not name a contrast, because there is no unique outcome associated with "i treated" once the rest of the village can change.

The project's kernel makes this concrete. For t = 0, …, T−1 with T = 20 and intervention date t* = 0,

$$Y_i(t+1) = \Lambda(\alpha + \beta\, Y_i(t) + \gamma\, (\Sigma_{j \text{ in } N(i)}\, Y_j(t))/k_i + \eta \cdot X_i + \kappa\, D_i + \varepsilon_i(t)),$$

where Λ is the logistic map, N = 800, the graph is a single Erdős–Rényi draw with mean degree four, and the parameter box sets γ in [0.05, 0.80]. The lower endpoint is positive on purpose: a no-interference world is not in the box (METHODOLOGY.md §2, §7). The peer term is a function of *neighbor outcomes*, which themselves depend on those neighbors' treatments and on *their* neighbors' outcomes. Interference is therefore not confined to the one-hop treatment vector. Over T periods a seed can move along paths. That is the scientific object. It is also why an estimator that treats (Y_i, D_i, X_i) as a SUTVA triple is answering a question the kernel does not pose.

Policy contrasts, not individual effects

Once potential outcomes are functions of **D**, well-posed estimands are comparisons of *assignment rules*. An assignment rule π is a (possibly degenerate) distribution on $\{0,1\}^N$. The associated mean outcome in world c is $E_c[\bar{Y}_T \mid \pi]$, the expectation of the terminal population mean over simulation noise ω — and, when π is random, over the draw of **D**. The project freezes three rules and three estimands (METHODOLOGY.md §3–4):

Object	Definition	Role
π_0	$D_i = 0$ for all i	Control rule
π_1	SRSWOR, k = 80 of N = 800	Random seed
π_2	$D_i = 1$ on the 80 highest-degree nodes (ties by index)	Hub seed; deterministic given A
$\theta_1(c)$	$E_c[\bar{Y}_T \mid \pi_1] - E_c[\bar{Y}_T \mid \pi_0]$	Random-seed contrast
$\theta_2(c)$	$E_c[\bar{Y}_T \mid \pi_2] - E_c[\bar{Y}_T \mid \pi_0]$	Headline
$\tau(c)$	$\theta_2(c) - \theta_1(c)$	Targeting premium

Table 1. Assignment rules and policy estimands. Expectations are over simulation noise; π_2 is a function of the realized graph, hence non-random given A.

These are comparisons of worlds under two complete assignments, not comparisons of treated units with untreated units inside one assignment. $\theta_2(c)$ asks: if we seed the hubs rather than seed nobody, what happens to mean terminal behavior in calibration c? That question remains well-defined when every unit interferes with every other unit. The individual contrast does not.

Direct, indirect, total, overall

A second, older vocabulary classifies contrasts by *whose* treatment is switched, holding a group-level program partly fixed. Halloran and Struchiner (1991, 1995), writing about vaccines and Ross's "dependent happenings," distinguished four effects. Hudgens and Halloran (2008) gave them potential-outcome definitions under partial interference (interference within groups, not across) and showed that, with two-stage randomization, they are unbiasedly estimable. In a group that faces coverage program α versus coverage program α':

- **Direct.** Switch i's own treatment, holding the others at a fixed program. Compare i treated under α with i untreated under the same α.
- **Indirect (spillover).** Hold i untreated and switch the program around i. Compare untreated units under α with untreated units under α'.
- **Total.** Switch both i's treatment and the program. Compare i treated under α with i untreated under α'. Hudgens and Halloran show that the total effect equals the sum of the direct and indirect effects.
- **Overall.** Compare the group mean under α with the group mean under α'. This is the closest analogue of θ_1 and θ_2: a contrast of assignment rules, averaging over treated and untreated units together.

Two comments, both load-bearing for the foils. First, a SUTVA estimator can speak only to a direct-effect analogue, and even that analogue is misnamed when interference is present: the coefficient on D_i in a regression of Y_i on D_i and X_i mixes own-treatment and the correlation between own treatment and neighbor treatment. Second, the project's headline θ_2 is an *overall* contrast of π_2 against π_0. Spillover is not a nuisance in that contrast; it is part of the policy. An estimator that cannot express spillover cannot be a competitor for θ_2. It can still be a foil — a method that a competent referee will propose, and that we therefore run and display.

2. A historical arc, not a succession

Three literatures are in play. They are parallel. The third did not supersede the first two.

Infectious-disease estimands, 1991–2008. Halloran and Struchiner (1991) constructed study designs that separate direct protection of the vaccinated from indirect protection of the unvaccinated, the latter arising because incidence depends on prevalence. Halloran and Struchiner (1995) translated the discussion into Rubin's potential-outcome notation and recorded the combinatorial explosion: with interference, the potential-outcome array is indexed by the full assignment. Hudgens and Halloran (2008), in JASA, produced the first clean randomization-based estimators of direct, indirect, total, and overall effects, under the restriction that the population partitions into groups with no cross-group interference and that both groups and individuals are randomized (two-stage randomization). The taxonomy in §1 is theirs. The project does not have two-stage randomization and does not have isolated groups. The vocabulary still applies; the 2008 estimators do not transplant.

Design-based exposure mappings, 2017–. Aronow and Samii (2017), in the *Annals of Applied Statistics*, dropped the partial-interference partition. Their framework has three pieces: (i) an experimental design, a known distribution on assignment vectors; (ii) an exposure mapping $g_i(\mathbf{D})$, a function that reduces the 2^N-valued argument to a low-dimensional exposure that the analyst is willing to treat as the relevant summary; (iii) estimands that are average contrasts of potential outcomes under two exposure values. Inverse-probability weighting (Horvitz–Thompson) and its ratio (Hájek) form deliver unbiasedness or approximate unbiasedness for those exposure contrasts whenever the design makes the exposure probabilities known and bounded away from zero. Sävje (2024) later separated the two roles exposure maps had been asked to play — defining the estimand, and correctly describing the interference structure — and gave conditions under which misspecified maps still yield precise estimates of the (now purely definitional) exposure contrast. Leung (2022) weakened exact neighborhood exclusion to *approximate* neighborhood interference, matching the spirit of a kernel in which effects decay with network distance but need not hit zero at distance one. Chin (2019) pursued a different reduction: regression adjustment for the *global* treatment effect in Bernoulli experiments, treating functions of **D** as features rather than coarsening **D** into strata. Forastiere, Airoldi, and Mealli (2021) took the exposure-mapping idea into observational networks, replacing known design probabilities with a generalized propensity score on own treatment and neighborhood treatment jointly.

Machine-learning heterogeneous treatment effects under SUTVA, 2016–. Separately, and not as a sequel, a literature developed estimators of $\tau(x) = \mu_1(x) - \mu_0(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$ from independent units. Athey and Imbens (2016) introduced honest recursive partitioning: one sample grows the tree, a disjoint sample estimates leaf effects, so that confidence intervals for leaf treatment effects need not pretend the partition was known. Wager and Athey (2018) lifted honesty to forests and proved pointwise consistency and asymptotic normality of the causal-forest estimator of $\tau(x)$ under unconfoundedness and SUTVA. Athey, Tibshirani, and Wager (2019) recast the forest as a locally weighted moment estimator (generalized random forests). Hahn, Murray, and Carvalho (2020) gave the Bayesian analogue, Bayesian causal forests (BCF), with a sum-of-trees prior on the prognostic function and a separately regularized prior on the treatment-effect function, and with an estimated propensity score included as a covariate to blunt regularization-induced confounding. Chipman, George, and McCulloch (2010) is the BART progenitor. All of this machinery estimates a SUTVA contrast, flexibly, with honest uncertainty. None of it is an interference estimator.

Parallel literatures

Causal forests did not replace exposure mappings, and exposure mappings did not replace the Halloran taxonomy. Forests are a leading method for heterogeneous individual effects under SUTVA. Exposure mappings are a leading method for design-based inference under a declared interference reduction. The project uses each as a foil for a different reason. Calling the forest "the modern estimator" in an interference problem is a category error.

3. Foil 1: causal forests and Bayesian causal forests

What is estimated

Write $\mu_d(x) = E[Y \mid X = x, D = d]$ and $\tau(x) = \mu_1(x) - \mu_0(x)$. Under SUTVA, unconfoundedness (D independent of $\{Y(0), Y(1)\}$ given X), and overlap, $\tau(x)$ equals the conditional average treatment effect $E[Y(1) - Y(0) \mid X = x]$. A causal forest estimates $\tau(x)$ by growing an ensemble of honest trees whose splitting criterion targets treatment-effect heterogeneity rather than prediction of Y (Wager and Athey 2018). The sample average of $\hat{\tau}(X_i)$, or a doubly robust aggregation of it, is then reported as an average effect. Bayesian causal forests (Hahn, Murray, and Carvalho 2020) instead specify

$$Y_i = \mu(X_i, \hat{\pi}(X_i)) + \tau(X_i)\, D_i + \varepsilon_i,$$

place a BART prior on the prognostic function μ and a second, more heavily regularized, BART prior on the treatment-effect function τ ("shrink toward homogeneity"), and include an estimated propensity $\hat{\pi}$ as an additional coordinate of μ so that the prior does not, through smoothness, reintroduce confounding that the propensity had removed. The output is a posterior over *functions* (μ, τ), given a dataset. It is not a posterior over calibrations c given anchors z. That distinction is the whole point of putting BCF in the foil slot. History matching reports an image of θ_2 over NROY. BCF reports a posterior for $\tau(\cdot)$ at a single c_0. Mixing the two as if they were interchangeable uncertainty statements is a mistake about the parameter.

Honesty

Honesty, in the Athey–Imbens–Wager sense, is a sample split inside the estimator. Tree structure (which coordinate to split, where to cut) is learned on one subsample; leaf estimates of μ_1 and μ_0 are formed on a disjoint subsample. Because the cutpoints are independent of the outcomes used to estimate effects, the leaf contrasts behave like pretabulated strata. Wager and Athey (2018) use that fact to construct pointwise confidence intervals for $\tau(x)$ that are centered at the forest and are asymptotically valid under their conditions. "Honest" here does not mean "covers $\theta_2(c)$ for c in NROY." It means "the interval for $\tau(x)$ does not ignore the search that produced the partition," *under SUTVA*. METHODOLOGY.md §9 asks for an honest interval from the forest. That is the right interval to request from this estimator. It is an interval for a SUTVA functional on one world.

Why the forest cannot see a γ edge

The simulated panel for foil 1 is one world c_0 (the median-implausibility survivor, once waves exist), with features X_i in $[0,1]^3$ (education, resource, baseline propensity), treatment D_i, and outcome s_i drawn from an admissible measurement m in $\{m_1$ oracle, m_2 survey$\}$. METHODOLOGY.md §9 takes D from π_2. The forest splits on coordinates of X. Neighbor treatment does not appear in X. Neighbor outcome does not appear in X. Degree does not appear in X. There is therefore no split that can isolate the peer term $\gamma\,(\Sigma_{j \text{ in } N(i)}\, Y_j)/k_i$. Adding

degree after the fact would still not identify γ: under π_2, D is a deterministic function of degree, so degree is the treatment, not a confounder one can partial out.

Even on π_1, where D is randomized and a SUTVA learner has a fighting chance at a direct-effect analogue, the estimand $\tau(x)$ is not $\theta_1(c)$. $\theta_1(c)$ is an overall contrast of two complete assignments, and it includes the equilibrium adjustment of untreated units. The forest's average of $\hat{\tau}(X_i)$ compares observed treated units with observed untreated units *inside the same assignment*, which is the contrast that Halloran and Struchiner would refuse to call overall, and that Hudgens and Halloran would refuse to call total. Architecture, not tuning, is the obstruction.

> **Not SOTA for interference**
>
> Causal forests and BCF are among the leading methods for heterogeneous individual effects under SUTVA, with a proof strategy (honesty, influence functions, Bayesian regularization of τ separate from μ) that deserves to be taught carefully. They are not state of the art for interference. Using them as a foil is a statement about architecture, not a claim that the authors of those papers missed a network trick.

Software, in one paragraph

Two Python libraries dominate applied causal ML, and they are not interchangeable. Microsoft's econml implements orthogonal / double machine learning estimators and honest-forest inference (including CausalForestDML), with interval estimators aimed at the SUTVA functionals those methods identify. Uber's causalml implements uplift and meta-learners (T-, S-, X-learners and related) aimed at ranking units for treatment at web scale. The diagnostic needs honest confidence intervals on a small simulated panel of N = 800, not a production uplift model. econml is the library that matches the inferential target; causalml is the library that does not. That is the entire comparison.

4. Foil 2: exposure-mapped design-based inference

AMENDMENT-01 (proposed 2026-08-25; **draft, not frozen**) adds a second diagnostic so that a referee cannot collapse two failures into one. Foil 1 fails because of SUTVA architecture *and* because it conditions on a single world. An exposure-mapped estimator of the Aronow–Samii type removes the first failure, on the arm where a design exists. Whatever gap then remains is not "you used the wrong estimator."

Design, map, estimand, weights

Aronow and Samii (2017) ask the analyst to name four objects.

1 **Design.** A known probability distribution on assignment vectors. Under π_1 this is simple random sampling without replacement of k = 80 treated nodes from N = 800. First-order inclusion probabilities are exact: $P(D_i = 1) = 80/800 = 1/10$. Second-order inclusions are exact: for $i \neq j$, $P(D_i = D_j = 1) = 80 \cdot 79 / (800 \cdot 799)$. Under π_2 there is no such distribution; see §6.

2 **Exposure map.** A function $g_i(\mathbf{D})$ that the analyst treats as the argument of i's potential outcome. The draft map is $g_i = (D_i, s_i)$ with $s_i = (\Sigma_{j \text{ in } N(i)} D_j)/k_i$, the share of treated neighbors ($k_i = \max(\text{degree}_i, 1)$). Continuous s_i is then coarsened, a priori, into four strata — not estimated from the outcome.

3 **Estimand.** A contrast of mean outcomes across exposure values. The draft reports two Hájek contrasts, defined below. These are exposure effects in Aronow–Samii's sense, not automatically equal to θ_1 or θ_2.

4 **Weights.** Inverse-probability weights for the event $\{g_i = d\}$. Horvitz–Thompson uses the raw inverse probabilities; Hájek normalizes them so the weights sum to one inside each exposure. The draft uses Hájek. Unbiasedness (HT) or consistency (Hájek) is with respect to the design, not with respect to a superpopulation, and it requires that every unit have a known, strictly positive probability of each exposure that the estimand invokes.

The frozen strata in the draft are

Stratum	D_i	s_i	Reading
S1	1	any	Treated
S0a	0	$s_i = 0$	Unexposed control
S0b	0	$0 < s_i < 0.25$	Lightly exposed untreated
S0c	0	$s_i \geq 0.25$	Heavily exposed untreated

Table 2. Draft exposure strata (AMENDMENT-01). Thresholds are declared before looking at Y. They are not a data-driven partition.

Hájek stratum means are the design-weighted averages of the observed outcome s (from m_1 or m_2; m_3 is excluded by the admission rule in METHODOLOGY.md §8) among units whose realized g_i falls in the stratum. The two contrasts are

$$\delta_{dir} = mean(S1) - mean(S0a),\ \delta_{spill} = mean(S0c) - mean(S0a).$$

δ_{dir} is a direct-effect analogue: treated versus unexposed untreated. δ_{spill} is a pure-spillover analogue: heavily exposed untreated versus unexposed untreated. A SUTVA estimator cannot even write δ_{spill}. That is why δ_{spill} is the overlay headline for foil 2. Variance is randomization-based, using the SRSWOR second-order inclusions, with a conservative variant preferred in the draft. Cell-size guard: if any stratum has fewer than 20 agents in a replication, merge S0b into S0c for *all* replications of that world and log the merge. The guard exists because mean degree is about four and only ten percent of nodes are treated; S0a (strictly unexposed) can be thin in a sparse graph, and empty cells make a Hájek contrast undefined.

What the map does not claim

The kernel transmits interference through neighbor *outcomes* over T periods, not solely through neighbor treatments at a single shot. The map $g_i = (D_i, s_i)$ is therefore a reduction, and in Sävje's (2024) terms it is a map one should be prepared to treat as misspecified relative to the true interference structure. The draft does not claim otherwise. An appendix robustness tick — recompute with threshold 0.5 and report whether the pre-registered F-* conclusions flip — is already declared. Forastiere, Airoldi, and Mealli (2021) would, in an observational network, additionally require unconfoundedness of own treatment *and* neighborhood treatment given individual and neighborhood covariates. Foil 2 does not take that road: on π_1 the design is known, so the probabilities are not estimated from X. That is the advantage of staying design-based. It is also the reason foil 2 cannot be moved onto π_2.

5. Worked example: a four-node path

AMENDMENT-01's pedagogical contract is a path graph on four nodes, A—B—C—D, with treatment set {B}. Compute s_i from the definition $s_i = (\Sigma_{j \text{ in } N(i)} D_j)/k_i$.

Node	Neighbors	k_i	D_i	Treated neighbors	s_i	Stratum
A	{B}	1	0	{B}	1/1 = 1	S0c ($s \geq 0.25$)
B	{A, C}	2	1	none	0/2 = 0	S1
C	{B, D}	2	0	{B}	1/2 = 0.5	S0c ($0.5 \geq 0.25$)
D	{C}	1	0	none	0/1 = 0	S0a

Table 3. Path A—B—C—D, treat {B}. Strata follow Table 2 and the definition of s_i. Cell counts here are all below the draft's guard of 20; the example exists to make the arithmetic inspectable, not to be a viable sample.

So s = (1, 0, 1/2, 0) for (A, B, C, D). A is untreated but fully exposed: its only neighbor is the treated node. D is the unique unexposed control. S0b is empty. S1 and S0a are singletons. A draft sentence in AMENDMENT-01 lists s = (0, 1/2, 1/2, 0) and maps A to S0a. That tuple is not the treated-neighbor share on this graph. Until the amendment is frozen, the formula in the same document is the authority, and Table 3 follows the formula. The qualitative lesson does not depend on the disputed coordinate: C is an untreated neighbor of a treated node with $s_C = 1/2 \geq 0.25$, hence S0c, and a merge fallback exists because strata this small cannot support a variance estimator.

Hájek contrasts on this toy assignment would be $\delta_{dir} = Y_B - Y_D$ and $\delta_{spill} = (Y_A + Y_C)/2 - Y_D$ (equal first-order weights inside a realized stratum of a complete enumeration; on the N = 800 graph the weights are the inverses of exposure probabilities, not uniform). With one unit in S0a the contrast is a number, not an estimate with a design-based interval. That is the point of the guard, and of not pretending a four-node path is the panel.

6. The design fact

π_2 is deterministic

Given the graph A, the rule π_2 always selects the same 80 nodes — the highest-degree agents, ties broken by index. There is no randomization mechanism and there are no unknown assignment probabilities to estimate. Design-based inverse-probability weights are therefore not defined for π_2: the realized assignment has probability one, every other assignment has probability zero, and untreated hubs have exposure probability zero (positivity fails). No Aronow–Samii estimator, and no Hájek contrast that requires known $\Pr(g_i = d)$, reaches the headline policy θ_2. Foil 2 lives on π_1. This is a result about targeted assignment, not a software limitation to be patched by pretending π_2 was Bernoulli.

The fact is stronger than “we chose not to randomize among hubs.” Many policy-relevant targeting rules are functions of a pre-treatment network: seed the highest-degree, seed a dominating set, seed estimated influencers. Each such rule, once the graph is fixed, is a degenerate design. The design-based toolkit that makes foil 2 honest on π_1 has nothing to bite on π_2. History matching does: $\theta_2(c)$ is well-defined for every c as a contrast of two simulation experiments, one under the hub rule and one under control, with no need for an assignment probability. That is one reason the identified set Θ_{hm} — the image of $\theta_2(c)$ for c in NROY — can speak to the headline when the exposure estimator cannot.

Comparison objects therefore do not share a horizontal axis. METHODOLOGY.md §7 reports Θ_{hm} as the empirical interval of θ_2 over NROY. The draft amendment reports Θ_{exp} as the image of $\delta_{spill}(c)$ for c in NROY. Each foil's interval is judged against its *own* honest set. The forest is compared with Θ_{hm} and with θ_{true} at c_0. E2 is compared with Θ_{exp} and with δ_{true} at c_0. Overlaying the forest interval on Θ_{exp}, or the E2 interval on Θ_{hm}, plots two different functionals and is not a finding.

7. Why two foils

A single foil confounds two failures. Failure A is architectural: the estimator's potential-outcome index set is smaller than the kernel's. Failure B is informational: the estimator conditions on one world c_0 and never sees NROY. A referee who is shown only the forest will say, correctly, that we used an estimator that cannot represent interference, and will not be forced to confront B. Adding foil 2 is the reply. E2 is an interference estimator. It still does not consume NROY. If, after waves exist, E2's interval sits inside Θ_{exp} and misses δ_{true} or the edges of Θ_{exp}, the remaining gap is calibration uncertainty. The sentence the paper wants available is: we fixed the estimator, and the gap remained.

Pre-registered findings in the draft (extension of METHODOLOGY.md §10) are therefore labelled by foil. F-OVERCONF-E2: I_{E2} lies in the interior of Θ_{exp} and/or excludes δ_{true}. F-HONEST-E2: I_{E2} covers Θ_{exp}. Analogous statements for the forest use Θ_{hm} and $\theta_{true}(c_0)$. None of these is a result yet. Waves have not been run. Reporting a winner would be fiction.

One validity gate is not a finding. At the oracle measurement m_1, where $s = Y_T$, if E2's point estimate misses $\delta_{true}(c_0)$ by more than its own confidence interval, the harness is broken: stop, fix, re-run. The exposure estimator, on the arm where the design exists and the outcome is not mismeasured, must recover a known contrast. That check tests code and algebra. It does not test calibration.

	Foil 1 (forest / BCF)	Foil 2 (exposure Hájek)
Arm	π_2 panel (METHODOLOGY §9)	π_1 only (draft)
Architecture	SUTVA: $\tau(x) = \mu_1(x) - \mu_0(x)$	$g_i = (D_i, s_i)$; δ_{dir}, δ_{spill}
Can write spillover?	No	Yes (δ_{spill})
Consumes NROY?	No (one world c_0)	No (one world c_0)
Honest set for the overlay	Θ_{hm} of θ_2	Θ_{exp} of δ_{spill}
Reaches headline θ_2?	No	No (wrong arm, wrong functional)
Positive control	Not a validity gate for θ_2	At m_1: miss beyond CI implies harness bug

Table 4. Planned diagnostic logic. No cell is a numerical result; waves have not been run. Amendment-01 is a draft.

8. What this diagnostic is not

This is not a criticism of Aronow–Samii or of BCF

Neither foil consumes NROY. That is an inputs limitation of the diagnostic, not a defect in the papers. Aronow and Samii (2017) give design-based inference for exposure contrasts when the design is known; they do not claim to propagate calibration uncertainty across a computer-experiment box. Hahn, Murray, and Carvalho (2020) give a posterior over functions given a dataset; they do not claim to give a posterior over calibrations given anchors. Running those estimators on one simulated world and comparing the resulting interval with an identified set is a statement about what the estimator is fed, not about whether the estimator does what its authors proved.

The diagnostic is also not a horse-race between "classical" and "modern" methods, not a demonstration that machine learning fails on networks, and not a substitute for the history-matching identified set. Foil 1 will produce a confident interval for the wrong functional on the right arm (π_2, the headline assignment) under the wrong architecture. Foil 2 will produce a design-based interval for a spillover analogue on the wrong arm (π_1, the only arm with a design). History matching remains the only procedure in the protocol that images the headline estimand θ_2 over the worlds the anchors have not ruled out. The foils exist to show that replacing that procedure with a more familiar estimator does not recover the missing coordinate of uncertainty — and, in the π_2 case, does not even reach the policy.

Status, restated so it cannot be skimmed past. METHODOLOGY.md §9 is frozen as a one-foil diagnostic on π_2. AMENDMENT-01 is a draft dated 2026-08-25; nothing in it is frozen until sign-off. No history-matching wave has been run. No forest has been fit. No Hájek contrast has been computed on the N = 800 panel. Table 3 is arithmetic on a four-node graph, not a simulation result. The oracle gate, the F-* labels, and the overlay figure are planned comparisons. They will be reported as findings only after the waves exist and the amendment, if accepted, is frozen into §9.

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